

A Parsimonious WNT–RA–HOX Plasticity Model Separates Stem-Cell Depletion from HOX Reprogramming

Complete 14-state system, mathematical analysis, global sensitivity, monostability check, and independently verified simulation study

Revised computational report — September 2026

Defensible central result

Under severe APC loss, the nominal 48-hour ATRA exposure reduces the total population by 4.3%, the absolute stem-like population by 30.4%, and the ALDH-like fraction by 21.0% relative to a matched untreated trajectory. Meanwhile, an exploratory reduced HOX-risk index increases from 0.286 to 0.433. The flux-based maintenance index is 0.314 at exposure end, below its exact contraction threshold of 0.5, and re-crosses that threshold at 94.9 h post-treatment-start (6.9 h after washout; 90% interval 94.4–95.6 h across a 160-sample Latin Hypercube parameter ensemble) while the stem-like burden remains 31.5% below control. Thus a favorable abundance response can coexist with both post-washout renewal pressure and an ATRA-induced HOXC-high/HOXA13-low transcriptional configuration — the same direction reported for an eight-gene HOX signature that predicts poor colorectal-cancer survival. The same 160-sample ensemble and its global-sensitivity ranking preserve this tension, an exhaustive multi-start search finds no bistability anywhere in the calibrated disease-severity range under either the cycle-averaged or the true periodic RA forcing, and the qualitative directions survive five alternative choices of the shared Hill exponent/threshold held fixed elsewhere in the model: the model’s population-level switch is implemented explicitly, not an emergent bifurcation. A numerical collinearity check further shows which specific parameter combinations are, and are not, separately constrained by these outputs. The model is calibrated for mechanistic consistency and directional comparison; it is not a fit to raw replicate data and is not clinically validated.

1 What changed, and why this revision

The model keeps the seven-state reduced molecular core and adds phenotype-conditioned HOX programs and cell populations, without unnecessary gene-by-gene dynamics.

1. Every molecular species is scaled by its own characteristic concentration; the common 1000 nM scale is not used.
2. The four aggregate HOX programs are dimensionless scores, not measured concentrations.
3. ATRA decreases the ALDH-like population, consistent with Osmond’s ALDEFLUOR result. HOXC reprogramming is distinct from ALDH abundance.
4. Total population N_T , absolute stem-like population N_S , stem-like fraction F_S , ALDH-like fraction F_A , and residual-stem composition S_A are reported separately. A fraction is never called a burden.
5. The former protein-ratio “stemness” score is replaced by a flux-derived maintenance number $\mathcal{R}_{SC}$ and bounded index $\mathcal{S}_{SC}$. Their thresholds are determined exactly by the sign of dN_S/dt .
6. TCF/LEF is not added as an ODE. Its reporter activity is not assumed to equal the modeled b concentration.
7. The eight-gene survival signature is not expanded into eight ODEs. The algebraic quantity I_H is explicitly a reduced HOX-risk index, not the clinical signature.
8. The informative combination, schedule, and duration plots are recomputed under the corrected model; none assumes that ATRA enriches the full-population ALDH-positive fraction.

This revision additionally: (9) corrects a diagram error in Fig. 1C, where HOXA5- and RA-driven differentiation were drawn reaching only the ALDH-like compartment although Eq. (23) gives both

stem-like compartments an identical functional dependence on h_5 and r ; (10) replaces the ten-parameter local perturbation screen with the same screen *plus* a rank-correlation (PRCC-style) global sensitivity analysis, so the qualitative claims in Section 10 rest on more than a single design point (Section 10.10); (11) adds an explicit multi-start equilibrium search of the fast molecular subsystem across the full disease-severity range, to check whether the model’s stem/non-stem switching is an emergent bifurcation or an explicit rule (Section 10.11); (12) adds a structural-identifiability discussion (Section 11); (13) grounds the risk index against the actual published eight-gene signature rather than a generic description of it (Section 10.4); (14) reports a dimensional equivalent for every reduced-model coefficient for which one is well defined, split by how the coefficient enters its equation (Tables 3–5); (15) repeats the multistability search under the model’s actual periodic RA forcing rather than only its cycle average, closing a scope limitation the previous check left open (end of Section 10.11); (16) adds a numerical practical- identifiability (sensitivity-collinearity) analysis that both confirms the structural argument above quantitatively and surfaces several additional non-identifiable parameter combinations it did not anticipate (Section 11.1); (17) replaces the plain-random uncertainty draw with true Latin Hypercube sampling, reports 5–95% intervals in place of raw min–max, and propagates the same ensemble through the flux-based maintenance-timing analysis to place a confidence interval on the exit-to-renewal re-crossing time (Sections 10.10, 10.3); and (18) checks whether the headline result is robust to the shared Hill-exponent/threshold simplification, holding every other parameter fixed (Section 10.12).

The resulting model has 14 states: seven molecular states, four aggregate HOX-program states, and three cell-population states.

2 State variables and notation

The complete state vector is

$$\mathbf{y} = (b, p, h_5, h_{13}, m, r, c, u_{BL}, u_{CL}, u_{BA}, u_{CA}, n_L, n_A, n_D)^T. \quad (1)$$

Table 1: All 14 dynamic state variables.

Module	Variables	Meaning
Molecular	$b, p, h_5, h_{13}, m, r, c$	Scaled β -catenin, APC, HOXA5, HOXA13, MYC, retinoic acid, and CYP26A1
HOX programs	u_{BL}, u_{CL}	Aggregate HOXB and HOXC scores in the LGR5-like phenotype
HOX programs	u_{BA}, u_{CA}	Aggregate HOXB and HOXC scores in the ALDH-like phenotype
Populations	n_L, n_A, n_D	LGR5-like, ALDH-like, and differentiated populations scaled by K_N

The parameter notation is modular and consistent:

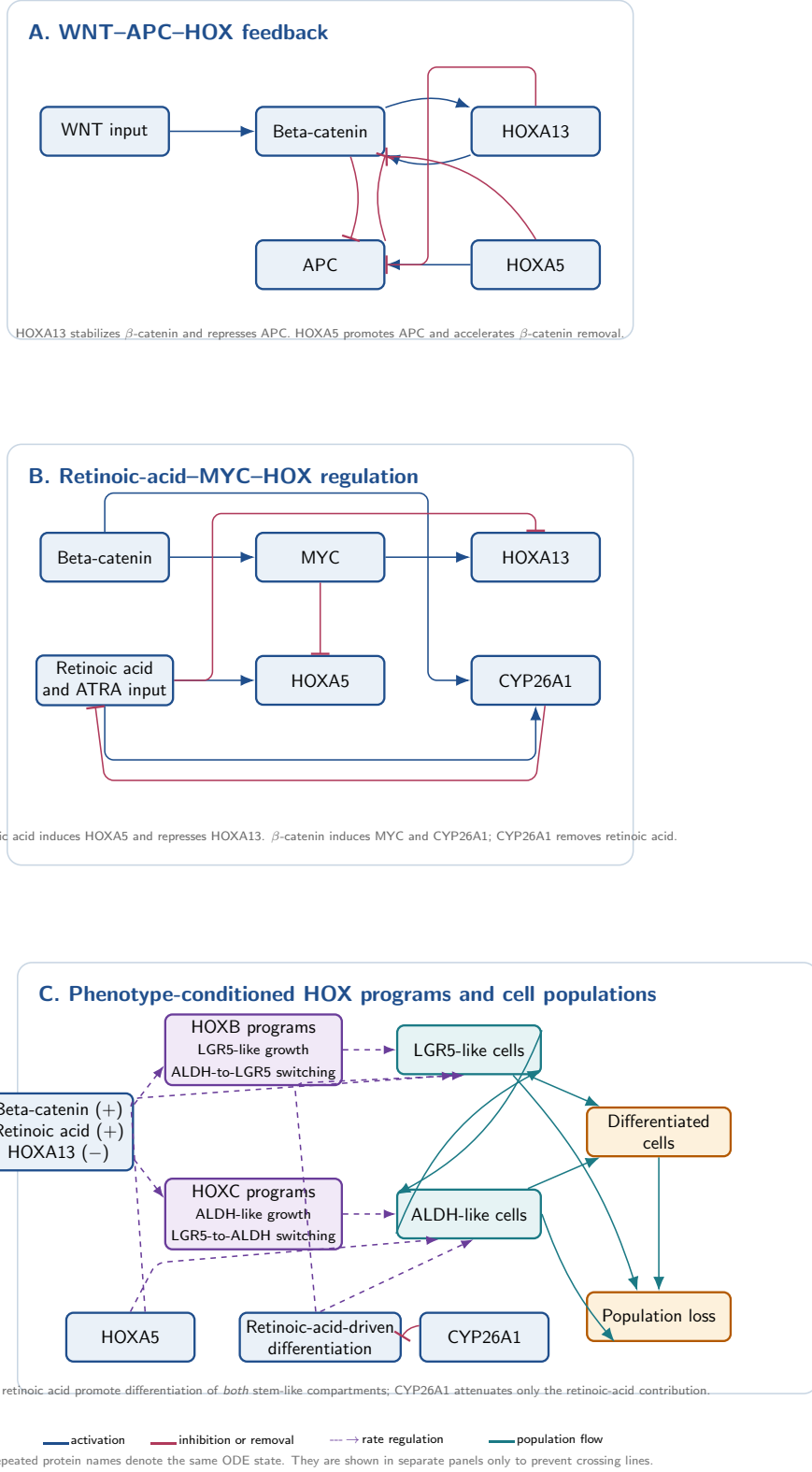


Figure 1: Complete regulatory architecture, organized into three panels to avoid crossed edges. Nodes use protein or population names only; mathematical symbols and parameter labels are intentionally omitted. Solid blue arrows are activation, red bar-headed edges are inhibition/removal, dashed violet arrows are rate regulation, and teal arrows are population flow. Repeated proteins are the same ODE state. Panel C is corrected from the previous version of this figure: HOXA5 and RA-driven differentiation now reach *both* LGR5-like and ALDH-like cells, matching the symmetric h_5 and r dependence that Eq. (23) gives to D_L and D_A alike.

Family	Meaning	Family	Meaning
s_X	basal source of X	v_{XY}	induction of X by Y
k_{XY}	removal/inhibition	w_{XY}	program weight
Γ_j	proliferative rate	Q_{ij}	phenotype switching
$\mathcal{D}_j$	differentiation rate	ε_X	relative time scale

3 Complete model schematic

Interaction audit: schematic lines versus equations

Every molecular line in Fig. 1 has exactly one modeled sign and an explicit location in the system:

Table 2: Molecular interaction audit. A bar denotes inhibition, repression, or enhanced removal.

Source $\rightarrow$ target	Modeled effect	Exact term or factor
WNT $\rightarrow$ β -catenin	activation	$+W$ in Eq. (6)
HOXA13 $\rightarrow$ β -catenin	stabilization	$+v_{b13}\mathcal{H}(h_{13})$ in Eq. (6)
APC $\dashv$ β -catenin	removal	$-k_{bp}pb$ in Eq. (6)
HOXA5 $\dashv$ β -catenin	removal	$-k_{b5}h_5b/(K+b)$ in Eq. (6)
HOXA5 $\rightarrow$ APC	activation	numerator $1 + a_{p5}h_5$ in Eq. (7)
β -catenin $\dashv$ APC	repression	denominator $1 + a_{pb}b + a_{p13}h_{13}$ in Eq. (7)
HOXA13 $\dashv$ APC	repression	same denominator in Eq. (7)
Retinoic acid $\rightarrow$ HOXA5	activation	$+v_{5r}\mathcal{H}(r)$ in Eq. (8)
MYC $\dashv$ HOXA5	removal	$-k_{5m}mh_5/(K+m)$ in Eq. (8)
β -catenin $\rightarrow$ HOXA13	activation	$+v_{13b}\mathcal{H}(b)$ in Eq. (9)
MYC $\rightarrow$ HOXA13	activation	$+v_{13m}\mathcal{H}(m)$ in Eq. (9)
Retinoic acid $\dashv$ HOXA13	repression	denominator $1 + q_{13r}\mathcal{H}(r)$ in Eq. (9)
β -catenin $\rightarrow$ MYC	activation	$+v_{mb}\mathcal{H}(b)$ in Eq. (10)
Retinoic acid $\rightarrow$ CYP26A1	activation	$+v_{cr}\mathcal{H}(r)$ in Eq. (12)
β -catenin $\rightarrow$ CYP26A1	activation	$+v_{cb}\mathcal{H}(b)$ in Eq. (12)
CYP26A1 $\dashv$ retinoic acid	removal	$-k_{rc}cr$ in Eq. (11)

The program/population lines are equally literal: β -catenin and retinoic acid raise, whereas HOXA13 lowers, both aggregate HOX-program drives; the LGR5-conditioned HOXB score regulates LGR5-like growth, the ALDH-conditioned HOXC score regulates ALDH-like growth, and the remaining two scores regulate the two switching directions. HOXA5 and retinoic acid promote differentiation of *both* stem-like states identically in functional form (Eq. (23)), CYP26A1 attenuates only the retinoic-acid contribution, both stem-like states flow to the differentiated state, and every population has explicit loss.

4 State-specific scaling

For dimensional concentrations $B, P, H_5, H_{13}, M, R, C$, define

$$B = B_0b, \quad P = P_0p, \quad H_5 = H_{5,0}h_5, \quad H_{13} = H_{13,0}h_{13}, \quad M = M_0m, \quad R = R_0r, \quad C = C_0c. \quad (2)$$

	B_0	P_0	$H_{5,0}$	$H_{13,0}$	M_0	R_0	C_0
reference (nM)	100	150	5	5	50	10	25

These are order-of-magnitude scaling references inherited from the reduced model, not universal measured abundances. The chosen β -catenin half-life fixes the time scale:

$$t_{1/2,B} = 50 \text{ min} = 5/6 \text{ h}, \quad d_B = \frac{\ln 2}{t_{1/2,B}} = 0.8317766 \text{ h}^{-1}. \quad (3)$$

For a species with degradation rate d_X , $\varepsilon_X = d_B/d_X$. The half-life choices 6, 6, 6, 0.5, 1, 4, and 8 h for APC, HOXA5, HOXA13, MYC, RA, CYP26A1, and the aggregate programs give

$$(\varepsilon_p, \varepsilon_5, \varepsilon_{13}, \varepsilon_m, \varepsilon_r, \varepsilon_c, \varepsilon_u) = (7.2, 7.2, 7.2, 0.6, 1.2, 4.8, 9.6). \quad (4)$$

Only the 50-minute half-life anchors the reference clock; the remaining half-lives are transparent plausible calibration choices. Cell populations satisfy $n_j = N_j/K_N$ with $K_N = 10^6$ cells. Population rates are scaled by $d_N = 1/24 \text{ h}^{-1}$, so $\varepsilon_N = d_B/d_N = 19.9626$.

For parsimony, all Hill functions use one reduced threshold and exponent:

$$\mathcal{H}(x) = \frac{x^n}{K^n + x^n}, \quad n = 2, \quad K = 0.50, \quad \mathcal{S}(z) = \frac{1}{1 + e^{-z}}. \quad (5)$$

The common *dimensionless* K does not imply one dimensional threshold: for $X = X_0 x$, $K_X = K X_0$. Thus the 0.5 thresholds are 50 nM for B , 75 nM for P , 2.5 nM for either HOX protein, 25 nM for MYC, 5 nM for RA, and 12.5 nM for CYP26A1.

5 Complete system: physical time, reduced concentrations

A prime denotes d/dt with t in hours throughout; every equation below is integrated exactly as written, and every rate constant is $d_X = d_B/\varepsilon_X$ so that this physical-time form is algebraically identical to the $\tau = d_B t$ rescaled system used to derive it. No asymptotic or quasi-steady-state argument is used anywhere in this report.

5.1 Seven molecular equations

$$b' = W + v_{b13}\mathcal{H}(h_{13}) - b - k_{bp}pb - k_{b5}\frac{h_5b}{K+b}, \quad (6)$$

$$p' = d_P \left[\frac{1 + a_{p5}h_5}{1 + a_{pb}b + a_{p13}h_{13}} - \Delta_P(\theta_P)p \right], \quad (7)$$

$$h_5' = d_5 \left[s_5 + v_{5r}\mathcal{H}(r) - h_5 - k_{5m}\frac{mh_5}{K+m} \right], \quad (8)$$

$$h_{13}' = d_{13} \left[\frac{s_{13} + v_{13b}\mathcal{H}(b) + v_{13m}\mathcal{H}(m)}{1 + q_{13r}\mathcal{H}(r)} - h_{13} \right], \quad (9)$$

$$m' = d_M [s_m + v_{mb}\mathcal{H}(b) - m], \quad (10)$$

$$r' = d_R [\mu_R(t) - r - k_{rc}cr], \quad (11)$$

$$c' = d_C [s_c + v_{cr}\mathcal{H}(r) + v_{cb}\mathcal{H}(b) - c], \quad (12)$$

where $\Delta_P(\theta_P) = 1 + \delta_P(1 - \theta_P)$, $\delta_P = 3.50$, and the stated rates are $d_B = 0.8318$, $d_P = d_5 = d_{13} = 0.1155$, $d_M = 1.3863$, $d_R = 0.6931$, $d_C = 0.1733$ (all h^{-1}); b itself carries d_B implicitly since $\varepsilon_b \equiv 1$ by construction. The smooth RA/ATRA input is

$$\mu_R(t) = \mu_0 + A_R \left[1 + \cos \left(\frac{2\pi t}{T_R} - \phi \right) \right] + \frac{D_R}{2} \sum_{k=1}^{n_w} a_k \left[\tanh\{q(t - t_k^{\text{on}})\} - \tanh\{q(t - t_k^{\text{off}})\} \right]. \quad (13)$$

The nominal experiment uses $n_w = 1$, $a_1 = 1$, $D_R = 1.50$, $q = 0.30 \text{ h}^{-1}$, $t^{\text{on}} = 40 \text{ h}$, $t^{\text{off}} = 88 \text{ h}$. D_R is a dimensionless intracellular input, not an extracellular μM concentration or a patient dose.

5.2 Four aggregate HOX-program equations

$$\Psi_B = w_{Bb}\mathcal{H}(b) + w_{Br}\mathcal{H}(r) - w_{B13}\mathcal{H}(h_{13}), \quad (14)$$

$$\Psi_C = w_{Cb}\mathcal{H}(b) + w_{Cr}\mathcal{H}(r) - w_{C13}\mathcal{H}(h_{13}), \quad (15)$$

$$u'_{BL} = d_U[\mathcal{S}(z_{BL} + \Psi_B) - u_{BL}], \quad u'_{BA} = d_U[\mathcal{S}(z_{BA} + \Psi_B) - u_{BA}], \quad (16)$$

$$u'_{CL} = d_U[\mathcal{S}(z_{CL} + \Psi_C) - u_{CL}], \quad u'_{CA} = d_U[\mathcal{S}(z_{CA} + \Psi_C) - u_{CA}]. \quad (17)$$

These are cluster-level program scores, not individual gene concentrations.

5.3 Three population equations

Let $N_T = n_L + n_A + n_D$. Then

$$n'_L = \Gamma_L n_L (1 - N_T) - (\mathcal{D}_L + \mu_L + Q_{LA})n_L + Q_{AL}n_A, \quad (18)$$

$$n'_A = \Gamma_A n_A (1 - N_T) - (\mathcal{D}_A + \mu_A + Q_{AL})n_A + Q_{LA}n_L, \quad (19)$$

$$n'_D = \mathcal{D}_L n_L + \mathcal{D}_A n_A - \mu_D n_D. \quad (20)$$

The rate functions are

$$\Gamma_L = \gamma_L^0 + \gamma_L^b \mathcal{H}(b) + \gamma_L^B \mathcal{H}(u_{BL}), \quad \Gamma_A = \gamma_A^0 + \gamma_A^b \mathcal{H}(b) + \gamma_A^C \mathcal{H}(u_{CA}), \quad (21)$$

$$Q_{LA} = q_{LA}^0 + q_{LA}^C \mathcal{H}(u_{CL}), \quad Q_{AL} = q_{AL}^0 + q_{AL}^B \mathcal{H}(u_{BA}), \quad (22)$$

$$\mathcal{D}_L = d_{L,0} + d_{L,5} \mathcal{H}(h_5) + \frac{d_{L,r} \mathcal{H}(r)}{1 + \chi_L \mathcal{H}(c)}, \quad \mathcal{D}_A = d_{A,0} + d_{A,5} \mathcal{H}(h_5) + \frac{d_{A,r} \mathcal{H}(r)}{1 + \chi_A \mathcal{H}(c)}. \quad (23)$$

Equations (18)–(23) are stated directly as physical-rate ODEs (all $\Gamma, Q, \mathcal{D}, \mu$ are h^{-1} , Table 6); dividing them by $d_N = 1/24 \text{ h}^{-1}$ recovers the dimensionless coefficients of the equivalent $\varepsilon_N n' = (\dots)$ form used to fix the population time scale, and the two forms integrate to the same trajectory. HOXA5 and retinoic acid enter $\mathcal{D}_L$ and $\mathcal{D}_A$ with the same functional form and only different numeric weights ($d_{L,5}$ vs. $d_{A,5}$, $d_{L,r}$ vs. $d_{A,r}$, χ_L vs. χ_A) — the structural symmetry that Fig. 1C now reflects correctly.

5.4 Outputs, not additional ODEs

$$N_S = n_L + n_A, \quad F_S = \frac{N_S}{N_T}, \quad F_A = \frac{n_A}{N_T}, \quad S_A = \frac{n_A}{N_S}. \quad (24)$$

N_T, N_S are abundances; F_S, F_A are fractions; S_A is residual-stem composition. For $N_S > 0$,

$$\bar{u}_C = \frac{n_L u_{CL} + n_A u_{CA}}{N_S}, \quad I_H = \frac{1}{2} [1 + \bar{u}_C - \mathcal{H}(h_{13})] \in [0, 1]. \quad (25)$$

Equal weights are used because gene-level coefficients are not identifiable (Section 11). I_H captures only the modeled HOXC-high/HOXA13-low direction and is distinct from the complete eight-gene survival signature (Section 10.4).

6 An interpretable stem-cell maintenance relationship

A molecular-ratio “stemness” score such as $b(1 + \alpha_{13}h_{13})/[(1 + p)(1 + \alpha_5h_5)]$ has the intended qualitative directions but no direct population-level meaning, and would count HOXA5 and HOXA13 a second time after their effects have already propagated through the molecular and population equations. We instead define a balance that follows exactly from the modeled population fluxes. Let

$$N_S = n_L + n_A, \quad E_L = \mathcal{D}_L + \mu_L, \quad E_A = \mathcal{D}_A + \mu_A. \quad (26)$$

Adding Eqs. (18)–(19) cancels both phenotype-switching terms and gives

$$N'_S = J_{\text{ren}} - J_{\text{exit}}, \quad J_{\text{ren}} = (1 - N_T)(\Gamma_L n_L + \Gamma_A n_A), \quad J_{\text{exit}} = E_L n_L + E_A n_A. \quad (27)$$

J_{ren} is realized production of stem-like cells through within-compartment proliferation, including density limitation; J_{exit} is removal through differentiation and cell loss. Switching changes LGR5-like/ALDH-like composition but does not change N_S directly. For $N_S > 0$, define the stem-cell maintenance number

$$\mathcal{R}_{\text{SC}} = \frac{J_{\text{ren}}}{J_{\text{exit}}}, \quad \mathcal{R}_{\text{SC}} > 1 \iff N'_S > 0, \quad \mathcal{R}_{\text{SC}} = 1 \iff N'_S = 0, \quad \mathcal{R}_{\text{SC}} < 1 \iff N'_S < 0. \quad (28)$$

A bounded version is preferable for figures:

$$\mathcal{S}_{\text{SC}} = \frac{\mathcal{R}_{\text{SC}}}{1 + \mathcal{R}_{\text{SC}}} = \frac{J_{\text{ren}}}{J_{\text{ren}} + J_{\text{exit}}} \in [0, 1], \quad N'_S = (J_{\text{ren}} + J_{\text{exit}})(2\mathcal{S}_{\text{SC}} - 1). \quad (29)$$

Its midpoint is not fitted or selected after simulation: $\mathcal{S}_{\text{SC}} = 0.5$ is exactly the boundary between contraction and expansion, and monotonicity is explicit: $\partial \mathcal{S}_{\text{SC}} / \partial J_{\text{ren}} = J_{\text{exit}} / (J_{\text{ren}} + J_{\text{exit}})^2 > 0$, $\partial \mathcal{S}_{\text{SC}} / \partial J_{\text{exit}} = -J_{\text{ren}} / (J_{\text{ren}} + J_{\text{exit}})^2 < 0$. Because the exact index includes crowding, it describes *realized* maintenance rather than a cell-autonomous molecular property. Separating the density contribution defines the uncrowded renewal potential

$$\mathcal{R}_{\text{SC}}^0 = \frac{\Gamma_L n_L + \Gamma_A n_A}{J_{\text{exit}}}, \quad \mathcal{R}_{\text{SC}} = (1 - N_T) \mathcal{R}_{\text{SC}}^0. \quad (30)$$

This construction embeds the relevant biology without adding fitted weights: Γ_L, Γ_A contain β -catenin and phenotype-conditioned HOX renewal signals, whereas $\mathcal{D}_L, \mathcal{D}_A$ contain HOXA5- and RA-driven differentiation attenuated by CYP26A1. APC and HOXA13 affect the index through the existing causal paths rather than through a second algebraic multiplier. Population-level stem-cell reproduction numbers have precedent in the APC-mutant colon model of Zhang et al. [7]; this formulation extends that idea to a dynamic, treatment-dependent WNT–RA–HOX system with two interconverting stem-like phenotypes. N_S , $\mathcal{S}_{\text{SC}}$, S_A , and I_H are retained separately because they respectively measure burden, maintenance pressure, phenotype composition, and a reduced transcriptional-risk direction.

7 Nominal parameter set

Every reduced-model coefficient below is dimensionless by construction (Section 4 non-dimensionalizes all seven molecular states by their own concentration scale X_0), but most coefficients still have a well-defined dimensional reading once the term they multiply is examined. Three cases arise, and Tables 3–5 are grouped accordingly. (i) A coefficient of a Hill-saturating production term $v_{XY}\mathcal{H}(Y)$

or a basal source s_X that is added directly into its *own* species' balance has dimensional equivalent (coefficient) $\times X_0$ in nM: this is the maximal (for v_{XY}) or basal (for s_X) concentration that term alone can contribute at steady state. (ii) A coefficient that multiplies another species' variable *linearly* or *bilinearly* (a repression-ratio term or a mass-action-type removal term) has dimensional equivalent (coefficient)/ Y_0 in nM^{-1} , the fractional change in the target's balance per nM of the regulating species Y ; Michaelis-type entries (k_{b5} , k_{5m}) report the low-concentration limiting sensitivity, since the exact sensitivity is state-dependent through the shared saturation constant K . (iii) A coefficient that acts only on a dimensionless HOX-program score $u_{BL}, u_{CL}, u_{BA}, u_{CA}$ (which have no concentration scale of their own), or that sets a fold-change inside a Hill-ratio denominator rather than an additive concentration, is dimensionless with no single-species concentration equivalent; these are listed with the term they shape instead of a converted value. The per-species dimensional Hill threshold $K_X = KX_0$ (Eq. (5)) is 50, 75, 2.5, 2.5, 25, 5, and 12.5 nM for $B, P, H_5, H_{13}, M, R, C$ respectively, as already noted in Section 4.

Table 3: Production, basal-source, and forcing parameters: each enters its own species' balance additively, so the dimensional equivalent is the coefficient times that species' own reference concentration X_0 (or, for μ_0, A_R , times R_0).

Parameter	Reduced value	Dimensional equivalent	Meaning
v_{b13}	0.90	90.0 nM	Max. HOXA13-driven β -catenin contribution
s_5	0.15	0.75 nM	Basal HOXA5 level
v_{5r}	2.50	12.5 nM	Max. RA-driven HOXA5 induction
s_{13}	0.18	0.90 nM	Basal HOXA13 level
v_{13b}	0.95	4.75 nM	Max. β -catenin-driven HOXA13 induction
v_{13m}	0.55	2.75 nM	Max. MYC-driven HOXA13 induction
s_m	0.18	9.0 nM	Basal MYC level
v_{mb}	1.35	67.5 nM	Max. β -catenin-driven MYC induction
s_c	0.08	2.0 nM	Basal CYP26A1 level
v_{cr}	1.50	37.5 nM	Max. RA-driven CYP26A1 induction
v_{cb}	1.50	37.5 nM	Max. β -catenin-driven CYP26A1 induction
μ_0	0.35	3.5 nM	Endogenous baseline RA level
A_R	0.04	0.4 nM	Amplitude of the daily endogenous RA oscillation
T_R	24 h (already dimensional)		Period of the endogenous RA oscillation

Table 4: Direct and Michaelis-type cross-coupling coefficients: each multiplies a *different* species' variable linearly or bilinearly, so the dimensional equivalent is the coefficient divided by that regulating species' own reference concentration, giving a per-nanomolar sensitivity.

Parameter	Reduced value	Dimensional equivalent	Meaning
a_{p5}	1.10	0.220 nM^{-1}	APC production sensitivity per nM HOXA5
a_{pb}	1.10	0.011 nM^{-1}	APC repression sensitivity per nM β -catenin
a_{p13}	1.30	0.260 nM^{-1}	APC repression sensitivity per nM HOXA13
k_{bp}	1.92	0.0128 nM^{-1}	β -catenin removal sensitivity per nM APC
k_{b5}	1.56	0.312 nM^{-1}	β -catenin removal sensitivity per nM HOXA5 (low- b limit)
k_{5m}	2.50	0.050 nM^{-1}	HOXA5 removal sensitivity per nM MYC (low- h_5 limit)
k_{rc}	0.85	0.034 nM^{-1}	RA removal sensitivity per nM CYP26A1

Table 5: Structural, fold-change, and HOX-program-score parameters: dimensionless by construction because they either shape the common Hill/logistic form itself, set a fold-change inside a ratio denominator, or act on a HOX-program score $u_{BL}, u_{CL}, u_{BA}, u_{CA}$ that carries no concentration scale. No single-species concentration equivalent applies.

Parameter	Reduced value	Role (no concentration equivalent)
K, n	0.50, 2	Common Hill threshold/exponent shape (dimensional thresholds per species listed above)
q_{13r}	1.00	Fold-strength of RA repression in the HOXA13 ratio denominator
χ_L, χ_A	1.50, 2.00	Fold-attenuation of CYP26A1 on RA-driven differentiation
z_{BL}, z_{BA}	0.50, -0.50	Logistic intercepts for the HOXB program in LGR5-like/ALDH-like cells
z_{CL}, z_{CA}	-0.50, 0.50	Logistic intercepts for the HOXC program in LGR5-like/ALDH-like cells
w_{Bb}, w_{Br}, w_{B13}	1.40, 0.35, 0.80	HOXB program weights on $\mathcal{H}(b), \mathcal{H}(r), \mathcal{H}(h_{13})$
w_{Cb}, w_{Cr}, w_{C13}	0.80, 1.60, 0.80	HOXC program weights on $\mathcal{H}(b), \mathcal{H}(r), \mathcal{H}(h_{13})$

Interpretation. The nM values in Table 3 are steady-state contributions, not rates: the physical production or removal *rate* for species X is this contribution times d_X (Table 6 gives the analogous rates already in h^{-1} for the cell-population equations). Every dimensional equivalent above is a reference-scale conversion of a calibrated reduced-model coefficient, not an independently measured biochemical rate constant; Section 11 discusses which combinations of these coefficients are, and are not, separately identifiable from the outputs reported here.

Table 6: Dimensional population-rate choices (h^{-1}).

Rate	h^{-1}	Rate	h^{-1}
$\gamma_L^0, \gamma_L^b, \gamma_L^B$	0.004, 0.016, 0.010	$\gamma_A^0, \gamma_A^b, \gamma_A^C$	0.003, 0.010, 0.012
q_{LA}^0, q_{LA}^C	0.0005, 0.0020	q_{AL}^0, q_{AL}^B	0.0010, 0.0080
$d_{L,0}, d_{L,5}, d_{L,r}$	0.0015, 0.012, 0.010	$d_{A,0}, d_{A,5}, d_{A,r}$	0.0015, 0.009, 0.015
μ_L, μ_A, μ_D	0.0010, 0.0015, 0.00722		

Table 7: Disease-progression regime coordinates.

Regime	Normal	Early adenoma	Advanced adenoma	Severe APC loss
W	0.80	1.00	1.50	2.00
θ_P	1.00	0.75	0.50	0.25

8 Positivity, boundedness, and well-posedness

Proposition 1. *For nonnegative initial molecular states, program scores in $[0, 1]^4$, and population states in $\Omega_N = \{(n_L, n_A, n_D) \in \mathbb{R}_+^3 : N_T \leq 1\}$, the system has a unique global solution that remains nonnegative and bounded.*

Proof. The vector field is locally Lipschitz on the nonnegative orthant because every denominator is strictly positive. On each molecular coordinate plane ($b = 0$, $p = 0$, etc.) the corresponding

derivative is nonnegative: a nonnegative source remains while every loss term containing that state vanishes. For a program score u , $d_U u' = d_U[\mathcal{S}(z) - u]$; at $u = 0$ the derivative is positive and at $u = 1$ it is nonpositive, so $[0, 1]^4$ is invariant. At $n_L = 0$, Eq. (18) leaves the nonnegative inflow $Q_{AL}n_A$; at $n_A = 0$, Eq. (19) leaves $Q_{LA}n_L$; and at $n_D = 0$, Eq. (20) leaves $\mathcal{D}_L n_L + \mathcal{D}_A n_A$. Therefore all populations remain nonnegative. Summing Eqs. (18)–(20) and evaluating on $N_T = 1$ gives $N_T' = -(\mu_L n_L + \mu_A n_A + \mu_D n_D) \leq 0$, so Ω_N is forward invariant: growth vanishes at $N_T = 1$ since every growth term carries a factor $(1 - N_T)$, switching cancels between the n_L and n_A equations, and differentiation cancels between loss from n_L, n_A and gain into n_D . For boundedness, use $0 \leq \mathcal{H}(x) \leq 1$ and the bounded ATRA input to bound every molecular variable except p by a linear comparison system; the resulting h_5 bound inserted into Eq. (7) gives a linear upper comparison for p as well. No finite-time blow-up is possible, so the local solution extends globally. $\square$

Numerically, every regime and condition tested has positive states, all four program scores remain in $[0, 1]$, and $N_T < 1$.

9 Simulation design

Each parameter set is preconditioned for 30 days without treatment to approach the untreated daily periodic state. Matched control and ATRA trajectories are then integrated for 168 h with a stiff BDF solver ($\text{rtol} = 10^{-9}$, $\text{atol} = 10^{-11}$, maximum step 0.5 h). Reported trajectories use a 0.125-h output grid. The primary exposure is 40–88 h; the end-of-treatment comparison is exactly 48 h after ATRA begins. Every rate constant in Section 5 is one of the nine physical numbers $d_B, d_P, d_5, d_{13}, d_M, d_R, d_C, d_U, d_N$; none is a formal small parameter and no asymptotic limit in any of them is taken anywhere in this report.

The common preconditioning seed, in the state order of Eq. (1), is

$$\mathbf{y}_{\text{seed}} = (0.20, 1.00, 0.80, 0.30, 0.30, 0.60, 0.40, 0.60, 0.40, 0.40, 0.60, 0.10, 0.15, 0.40)^T. \quad (31)$$

It initializes only the 720-h untreated preconditioning interval; treatment simulations begin from the resulting regime-specific periodic state. The severe-APC-loss calibration was constrained to reproduce directions in the Osmond studies: modest proliferation reduction under ATRA alone, an approximately 20% decrease in the ALDH-positive fraction, HOXA5 induction, later-HOX/HOXA13 reduction, CYP26A1 induction, and an aggregate HOXC-high/HOXA13-low response. Because replicate-level data were not fitted, agreement is calibration consistency, not independent validation.

10 Results

10.1 Molecular response

At 88 h, the treated/control ratios are

$$b : 0.719, \quad p : 1.890, \quad h_5 : 4.116, \quad h_{13} : 0.677, \quad m : 0.965, \quad r : 2.714, \quad c : 1.389.$$

The CYP26A1 direction matches experiment, but its 1.39-fold modeled protein/activity response is not a quantitative reproduction of the reported ~ 30 -fold mRNA induction reported for CYP26A1 in APC-mutant CRC [2].

Molecular response to a 48-hour ATRA exposure under severe APC loss

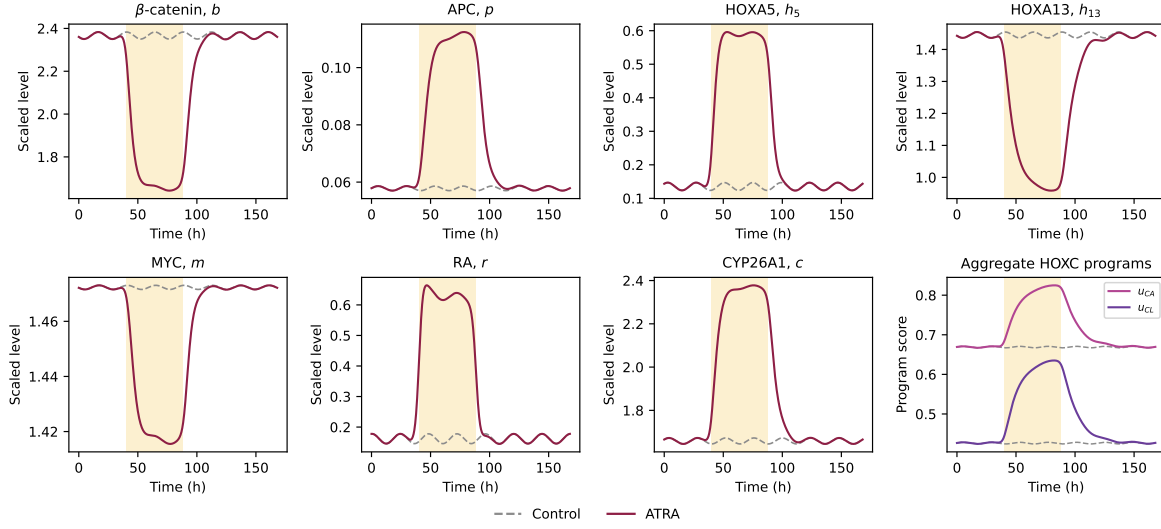


Figure 2: Nominal response under severe APC loss. ATRA raises RA, HOXA5, APC, CYP26A1, and aggregate HOXC programs while lowering β -catenin, HOXA13, and MYC. Smooth onset and washout come from the exposure function, not post-processing.

ATRA lowers stem-like populations and transiently raises the HOX-risk index

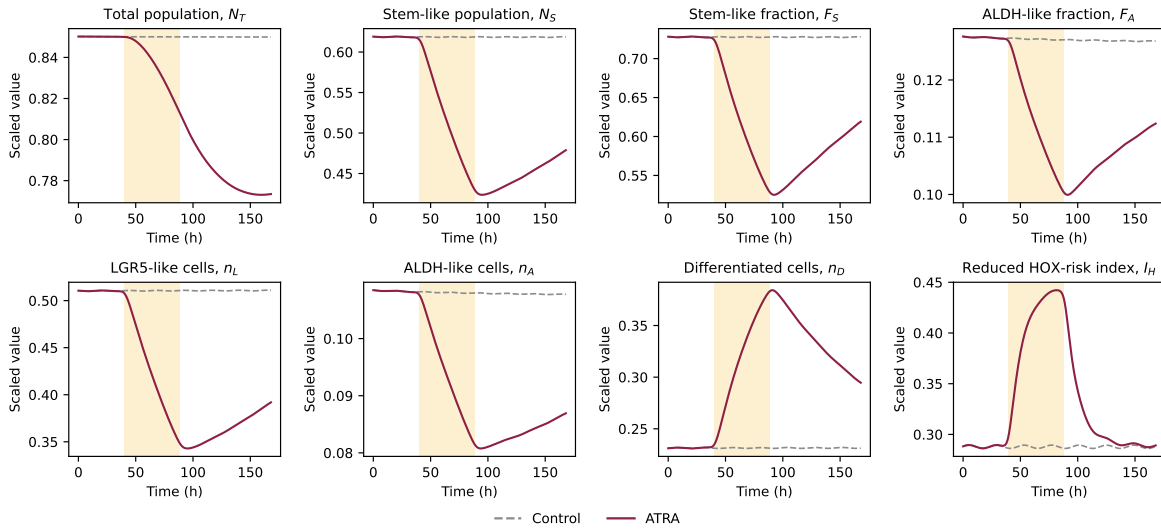


Figure 3: ATRA lowers both stem-like populations, raises the differentiated state, and transiently increases the reduced HOX-risk index.

Table 8: End-of-exposure response at 88 h. Ratios compare ATRA with matched control within each disease regime.

Regime	N_T ratio	N_S ratio	F_S ratio	F_A ratio	I_H ratio
Normal	0.923	0.617	0.669	0.757	2.156
Early adenoma	0.943	0.663	0.703	0.777	1.661
Advanced adenoma	0.954	0.688	0.722	0.787	1.541
Severe APC loss	0.957	0.696	0.728	0.790	1.514

10.2 Population response and HOX-program tension

In severe APC loss, N_T changes from 0.8500 to 0.8134, N_S from 0.6188 to 0.4309, and F_A from 0.1271 to 0.1003. The absolute ALDH-like population falls 24.4%. Since LGR5-like cells fall more strongly, the ALDH-like share within the residual stem pool rises modestly from 0.1746 to 0.1894 (+1.48 percentage points). This is a prediction about residual composition, not the experimental ALDEFLUOR endpoint.

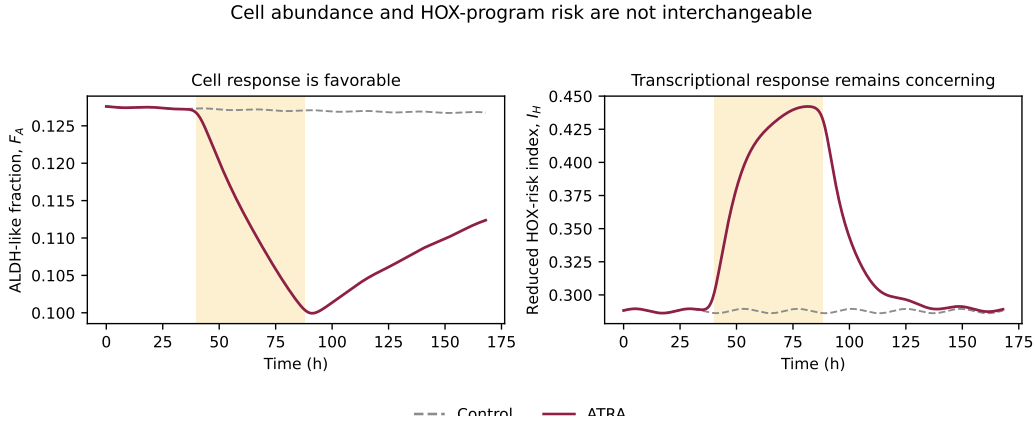


Figure 4: The central hypothesis: the ALDH-like fraction can decrease while the reduced HOX-risk index increases. The prognostic signature is associative; this does not show that ATRA causes worse survival.

10.3 Dynamic stem-cell maintenance and post-washout rebound

Table 9: Nominal severe-APC stem-maintenance results.

Event	Time (h)	N_S ratio	$\mathcal{S}_{SC}$	$\mathcal{R}_{SC}$	g_{SC} (h^{-1})
Minimum maintenance index	50.5	0.928	0.241	0.317	-8.18×10^{-3}
End of exposure	88.0	0.696	0.314	0.457	-5.52×10^{-3}
Threshold re-crossing	94.9	0.685	0.500	1.000	≈ 0
24 h after washout	112.0	0.699	0.595	1.468	1.70×10^{-3}
80 h after washout	168.0	0.774	0.591	1.443	1.75×10^{-3}

At 88 h the matched control has $\mathcal{S}_{SC} = 0.511$, whereas ATRA gives 0.314. The index re-crosses 0.5 about 6.9 h after nominal washout, and the simulated N_S minimum occurs at 94.9 h. Their

ATRA causes exit-dominated depletion followed by renewal-dominated rebound

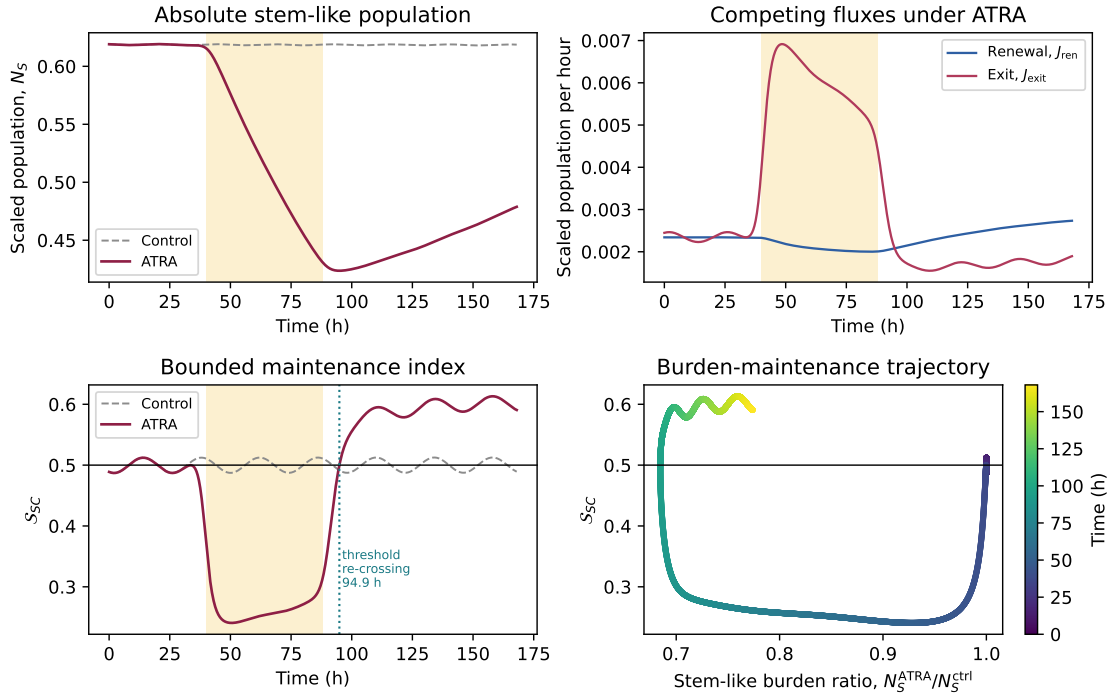


Figure 5: Flux-based interpretation of the nominal severe-APC-loss response. ATRA makes exit exceed renewal during exposure, forcing $S_{SC} < 0.5$ and decreasing N_S . After washout the fluxes cross at 94.9 h, almost exactly at the stem-like-population nadir. Maintenance becomes renewal-dominated while the absolute stem-like burden remains below its matched control.

agreement is required by Eq. (29), not an independent fitted coincidence. At 168 h the stem-like population is still 22.6% below control, but $\mathcal{R}_{\text{SC}} = 1.443$: the residual pool is smaller yet renewal-dominated. This is model-defined latent rebound pressure rather than evidence of clinical recurrence. The decomposition in Eq. (27) clarifies the source of rebound: at 88 h, $\mathcal{R}_{\text{SC}}^0 = 2.452$ but crowding and treatment-dependent exit reduce the realized number to 0.457; at 168 h, $\mathcal{R}_{\text{SC}}^0 = 6.369$ and the lower total density gives $1 - N_T = 0.227$, producing $\mathcal{R}_{\text{SC}} = 1.443$. The rebound combines recovery of the renewal-to-exit balance with ecological release from density limitation.

Table 9 reports these event times at the nominal parameter set only. To know whether the 94.9 h re-crossing time is a fragile point estimate or a robust feature of the model, we recomputed it on every one of the 160 Latin Hypercube parameter draws of Section 10.10 (same $\pm 15\%$ perturbations of the ten most influential couplings), each propagated through a full 168-h trajectory at the 0.125-h output grid of Section 9 rather than through the three-point snapshot used for the ratio outputs. The re-crossing time is tightly constrained: median 94.9 h with a 5–95% interval of [94.4, 95.6] h,

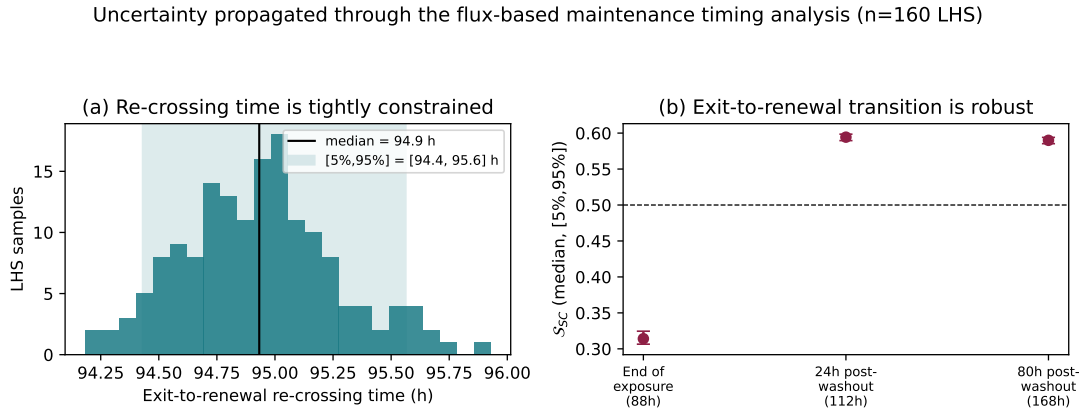


Figure 6: Uncertainty propagated through the flux-based maintenance timing analysis across the same 160-sample LHS ensemble as Fig. 14. (a) The exit-to-renewal re-crossing time is tightly constrained: median 94.9 h, 5–95% interval [94.4, 95.6] h. (b) The exit-dominated (88 h) to renewal-dominated (112 h, 168 h) transition itself never fails to occur in any sample.

i.e. within about 40 minutes of the nominal value across the full perturbation ensemble. The stem-like-population nadir tracks it just as closely (median 94.9 h, interval [94.4, 95.6] h), and the nadir-to-control ratio has interval [0.660, 0.712] (nominal 0.696, Table 9). Every one of the 160 samples reproduces the same qualitative transition — exit-dominated at 88 h, renewal-dominated by 112 h and 168 h — with no sample crossing back the other way. The headline claim in the central-result summary is therefore not a fitted coincidence at one parameter point: it is a robust, narrowly-bounded feature of the calibrated model structure.

10.4 Directional comparison with Osmond

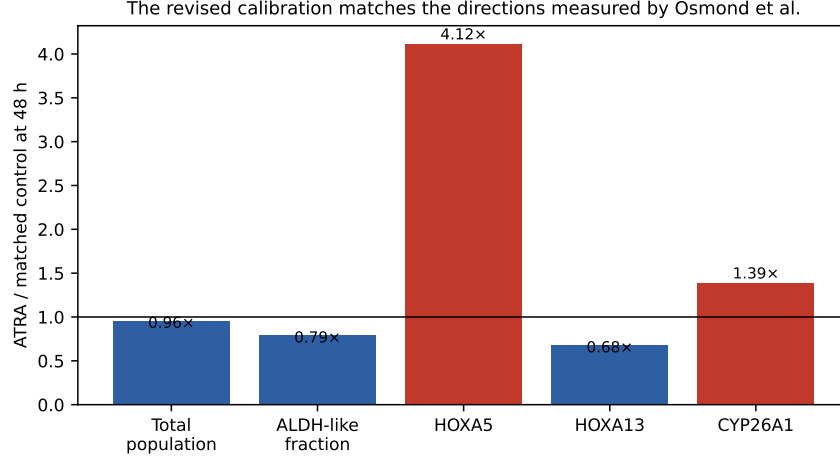


Figure 7: Direction-level comparison at 48 h. This is calibration consistency, not independent validation.

Table 10: Evidence audit for the revised model.

Experimental observation	Model result	Status
ATRA modestly reduces proliferation in mutant-APC HT29 cells [1, 3]	N_T decreases 4.3%	Directionally consistent; no direct cell-count fit
ATRA decreases the ALDH-positive fraction about 20% [3]	F_A decreases 21.0%	Calibration target reproduced; raw-data fitting is still required
ATRA increases HOXA5 and decreases later HOX genes including HOXA13 [3, 4]	h_5 rises 4.12 $\times$; h_{13} falls to 0.677 $\times$	Directionally consistent
ATRA strongly induces CYP26A1 mRNA [2]	c rises 1.39 $\times$	Direction only; magnitude and molecular observable differ
ATRA can increase TCF/LEF reporter activity in mutant-APC cells [2]	b decreases	Reporter activity and concentration differ; reporter is outside the ODE core
ATRA's HOX pattern matches an eight-gene poor-survival signature (HOXB4, HOXC6, HOXC9, HOXC11, HOXC13, HOXD9 up; HOXA13, HOXB13 down) [4]	I_H rises	Partial structural correspondence on the HOXC-up/HOXA13-down axis; I_H is not the eight-gene signature

10.5 Dose–duration response

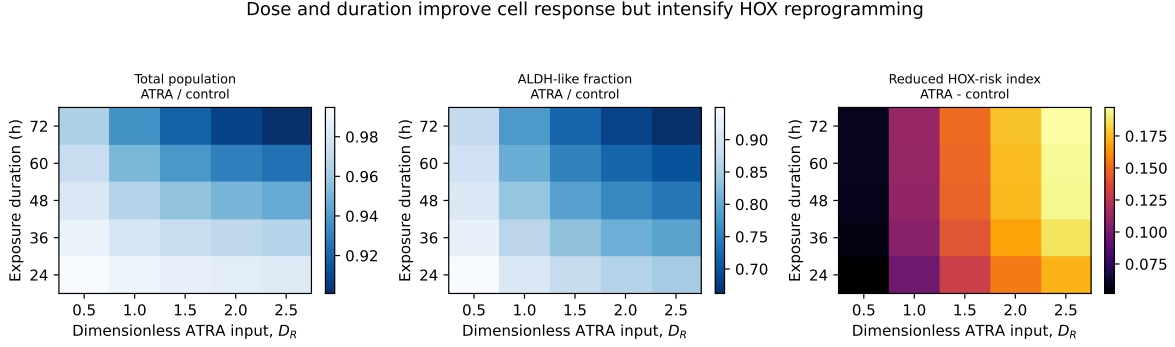


Figure 8: Severe-APC-loss scan at the end of exposure. Increasing D_R or duration improves total and ALDH-like responses but also increases I_H . D_R is dimensionless and is not a patient dose.

At $D_R = 1.5$, extending exposure from 24 to 72 h changes the total-population ratio from 0.987 to 0.919 and the ALDH-fraction ratio from 0.880 to 0.721. The HOX-risk increase changes from 0.130 to 0.150 and begins to saturate. The 48-hour window remains primary because it matches the experimental design and avoids presenting longer exposure as an unqualified improvement.

Maintenance transition across the dose–duration grid

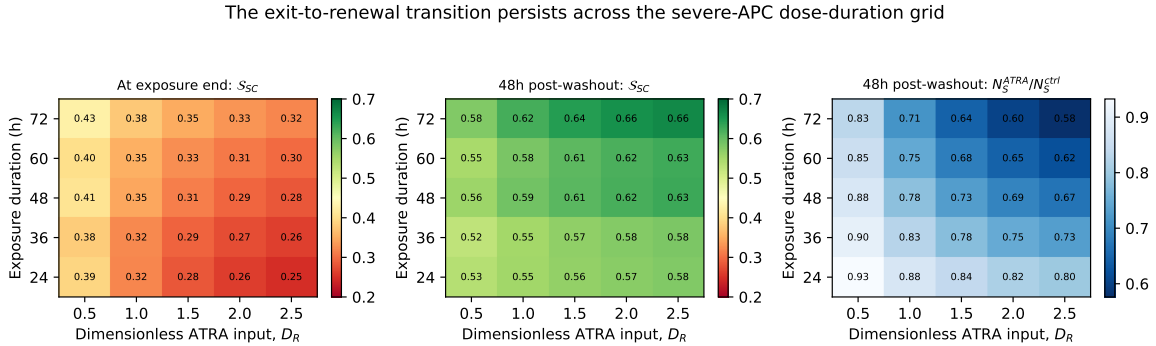


Figure 9: Every condition is exit-dominated at its exposure endpoint ($S_{SC} < 0.5$), but renewal-dominated 48 h after washout ($S_{SC} > 0.5$). The stem-like burden nevertheless remains below the matched untreated value.

Across the grid, the end-of-exposure maintenance index ranges from 0.249 to 0.428. Forty-eight hours after washout it ranges from 0.524 to 0.664, while the corresponding stem-like-burden ratio remains between 0.576 and 0.933. The exit-to-renewal transition is therefore not unique to the nominal $D_R = 1.5$, 48-h choice. This is protocol robustness within the stated grid; it does not establish behavior outside the calibrated parameter range.

10.6 Smooth exposure-duration trajectories

At the common 168-h follow-up, the total-population ratios for the 24, 48, 60, and 72-h exposures are 0.955, 0.910, 0.889, and 0.869; the ALDH-fraction ratios are 0.947, 0.886, 0.853, and 0.816. The

Exposure duration controls response depth and recovery

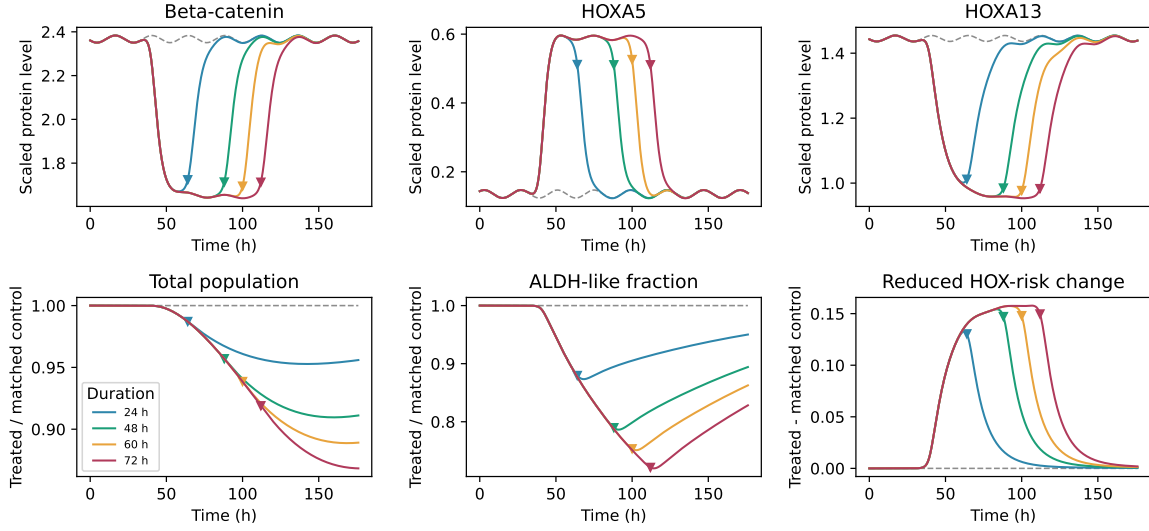


Figure 10: Fixed-amplitude exposures lasting 24, 48, 60, or 72 h under severe APC loss. Longer exposure deepens the cell response and prolongs recovery, whereas the HOX-risk increase approaches a plateau during treatment and decays after washout.

Table 11: Duration comparison at each schedule's end of exposure. Input amplitude fixed at $D_R = 1.50$.

Duration	Assessment time	N_T ratio	F_A ratio	ΔI_H
24 h	64 h	0.987	0.880	0.130
48 h	88 h	0.957	0.790	0.147
60 h	100 h	0.939	0.753	0.148
72 h	112 h	0.919	0.721	0.150

risk-index changes have largely washed out (0.0004–0.0026), whereas population effects persist. The 48-h case remains the primary evidence-linked condition; longer windows are model predictions for prospective testing.

10.7 Equal-input schedule comparison

The three schedules below have the same nominal added-input area, $D_R \sum_k a_k(t_k^{\text{off}} - t_k^{\text{on}}) = 72$ dimensionless-hours: continuous input from 40–88 h, a twofold front-loaded input from 40–64 h, and four twofold 6-h pulses distributed between 40 and 88 h. At 168 h, continuous, front-loaded, and four-pulse schedules give N_T ratios 0.910, 0.937, and 0.913 and F_A ratios 0.886, 0.926, and 0.889. Input area alone is insufficient: spacing and the time elapsed since the last exposure change the apparent response. It does not identify a clinically optimal schedule. The oscillation visible in the middle panel of Fig. 11 is not numerical noise: it reflects the fast RA/CYP26A1 incoherent feed-forward loop (RA directly induces HOXA5 within about an hour, but also induces its own clearance enzyme CYP26A1 on a slower, roughly 4-h time scale) sitting underneath the slower phenotype-program and population relaxation (roughly 11-h and 17-h time scales respectively); because every

Schedule changes response composition even at equal nominal ATRA input

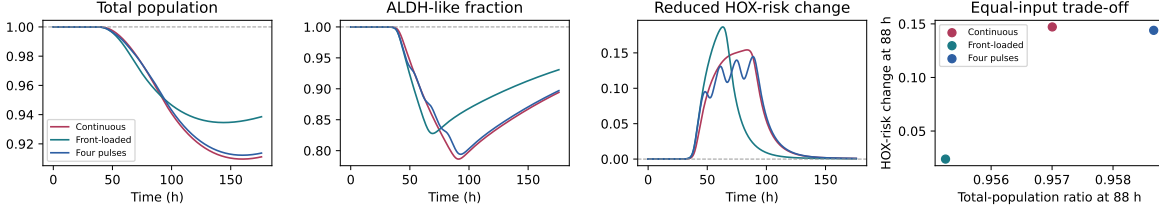


Figure 11: Equal nominal input does not imply equal dynamics. At the common 88-h comparison, front-loading has the smallest residual HOX-risk shift but a weaker ALDH-fraction response; continuous and pulsed exposure produce similar late risk shifts but different transient paths. The right panel is an endpoint trade-off, not an optimization frontier.

Table 12: Equal-input schedules at the common 88-h endpoint.

Schedule	Input area	N_T ratio	F_A ratio	ΔI_H
Continuous	72	0.957	0.790	0.147
Front-loaded	72	0.955	0.854	0.024
Four pulses	72	0.959	0.804	0.144

schedule shows the same two-time-scale overshoot, the effect is intrinsic to the circuit and not an artifact of discrete dosing.

10.8 Counterfactual selective HOX-risk-arm blockade

To recover a useful combination-design analysis without reintroducing an unsupported ATRA-to-ALDH-enrichment route, define a concurrent block fraction $\rho_H \in [0, 1]$ and the smooth exposure gate $G(t) \in [0, 1]$ already implicit in Eq. (13). Only two hypothesized risk-arm couplings are modified:

$$q_{13r} \mapsto [1 - \rho_H G(t)] q_{13r}, \quad w_{Cr} \mapsto [1 - \rho_H G(t)] w_{Cr}. \quad (32)$$

The perturbation attenuates retinoic-acid repression of HOXA13 and retinoic-acid activation of the HOXC program during exposure; it leaves the RA state, HOXA5 induction, and RA/HOXA5-driven differentiation intact. The near-identical N_T curves are informative rather than a failure: in

Table 13: Combination outcomes at the 88-h end of exposure.

Condition	N_T ratio	F_A ratio	ΔI_H
ATRA	0.9570	0.7897	0.1472
ATRA + 80% risk-arm block	0.9570	0.7605	0.0217
ATRA + complete risk-arm block	0.9570	0.7346	-0.0102

this model, the block is orthogonal to the dominant RA/HOXA5 differentiation route. The stronger ALDH-fraction response arises because HOXC-linked LGR5-to-ALDH switching is reduced. This is a sharp experimental discriminator: if a selective HOX perturbation changes total killing strongly, the proposed mechanism is incomplete. No existing compound is claimed to implement Eq. (32).

Selective HOX-risk-arm blockade preserves the ATRA cell response

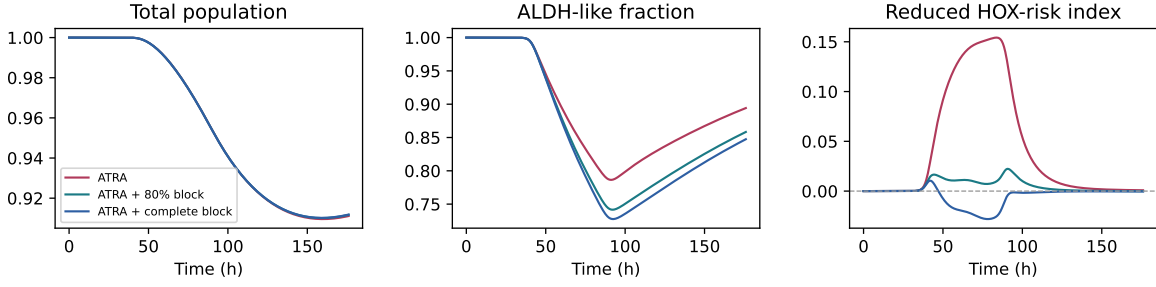


Figure 12: Theoretical combination experiment. Selectively weakening the RA-to-HOXC/HOXA13 risk arm leaves the total-population response almost unchanged and further decreases the modeled ALDH-like fraction, while suppressing the transient HOX-risk increase. “Complete block” is a mechanism probe, not a proposed drug effect.

10.9 Mechanism ablations

The three ablations use the following exact replacements:

$$\text{No RA/HOXA5 differentiation: } d_{L,5} = d_{A,5} = d_{L,r} = d_{A,r} = 0.$$

$$\text{No RA-to-HOX reprogramming: } v_{5r} = q_{13r} = w_{Br} = w_{Cr} = 0.$$

$$\text{No CYP26A1 attenuation: } \chi_L = \chi_A = 0.$$

The full model gives $F_A^{\text{ATRA}}/F_A^{\text{ctrl}} = 0.790$. With RA/HOXA5-driven differentiation removed,

Mechanism ablations separate cell and transcriptional responses

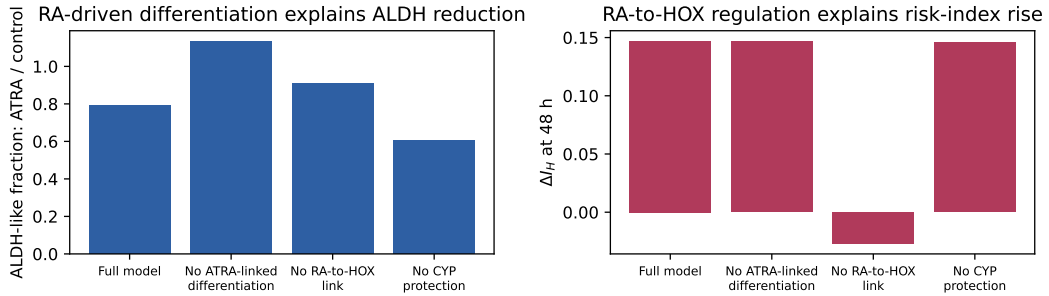


Figure 13: Removing all RA/HOXA5-linked differentiation eliminates the ALDH-fraction reduction (it overshoots past 1); removing RA-to-HOX regulation eliminates the I_H increase and turns it slightly negative; removing CYP26A1 attenuation deepens the cell response further. These ablations identify which assumed mechanism generates each result; they do not establish those mechanisms experimentally.

the ratio becomes 1.133 (ATRA now modestly *increases* the ALDH-like fraction, since the only remaining route out of the stem compartments is the small basal rate). With RA-to-HOX reprogramming removed, ΔI_H changes from +0.147 to -0.027 : the residual small negative value reflects

the indirect route by which reduced WNT/ β -catenin tone still shifts the HOXB/HOXC program weights, even with the direct RA-to-HOX terms zeroed.

10.10 Local and global sensitivity

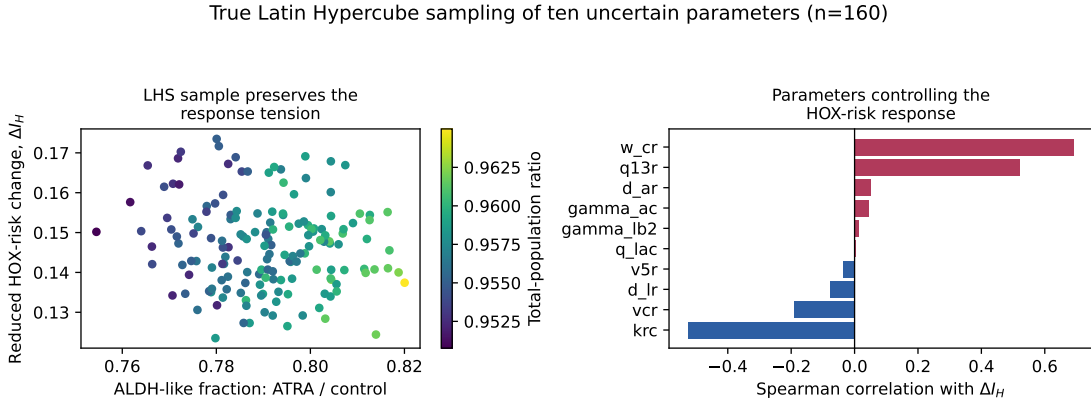


Figure 14: True Latin Hypercube sampling (`scipy.stats.qmc`), the sampling design recommended by Marino et al. [8] for this diagnostic, of ten uncertain parameters, $\pm 15\%$ log-uniform ($n = 160$). Every sample has $F_A^{\text{ATRA}}/F_A^{\text{ctrl}} < 1$ and $\Delta I_H > 0$. The right panel ranks each factor’s Spearman rank correlation with ΔI_H across the same 160 samples — a partial-rank-correlation-style global sensitivity screen, not a restatement of the local scatter.

This revision replaces the previous independent random log-uniform draw with a true Latin Hypercube sample (LHS), which fills the ten-dimensional perturbation space more evenly for the same sample budget and is the design Marino et al. [8] themselves recommend pairing with PRCC. Across the 160 LHS samples, the total-population ratio has a 5–95% interval of $[0.953, 0.961]$ (median 0.957; full range $[0.951, 0.965]$), the stem-population ratio $[0.675, 0.722]$ (median 0.698), the ALDH-fraction ratio $[0.770, 0.813]$ (median 0.791), and $\Delta I_H \in [0.130, 0.167]$ (median 0.147). All samples also preserve HOXA5 up and HOXA13 down. We report the 5–95% interval rather than the raw sample min–max, since the latter is sensitive to sample count in a way the interval is not; this is local robustness within the proposed structure, not global validation or posterior uncertainty.

Because a one-at-a-time $\pm 15\%$ screen can miss parameter combinations that push the model further from its nominal behavior, we additionally rank all ten factors by their Spearman rank correlation with ΔI_H across the same 160-sample LHS draw (a partial-rank-correlation, or PRCC, style diagnostic standard in systems-biology sensitivity analysis [8]). The RA-to-HOXC weight w_{Cr} ($\rho = 0.69$) and the RA-represses-HOXA13 strength q_{13r} ($\rho = 0.52$) are the dominant positive drivers of the risk index; CYP-mediated RA clearance k_{rc} ($\rho = -0.53$) and RA-induces-CYP26A1 strength v_{cr} ($\rho = -0.19$) are the dominant negative drivers. This is the same qualitative ranking implied by the mechanism ablations in the previous section (the RA-to-HOX arm and its CYP26A1 negative feedback are what move I_H), now shown to hold across correlated variation in all ten uncertain couplings simultaneously rather than only at the ten single-parameter perturbations; the ranking is stable under the LHS/PRCC upgrade even though the individual ρ values shifted somewhat from the earlier plain-random draw. The flux-based conclusion is also preserved in all 160 samples: at 88 h, $\mathcal{S}_{SC}$ has 5–95% interval $[0.307, 0.325]$, so every perturbed system is exit-dominated; at 112 h the interval is $[0.590, 0.599]$ and at 168 h $[0.585, 0.594]$, so every sample is renewal-dominated after

washout. We take this LHS ensemble further in Section 10.3 (Fig. 6) to place a confidence interval on the exit-to-renewal re-crossing time itself, not only on the endpoint ratios.

10.11 Multistability check of the fast molecular subsystem

A stem/non-stem “switch” can arise from two structurally different sources: an emergent bifurcation in the fast molecular circuit (bistability, so that two stable transcriptional states coexist for the same W, θ_P and the system’s history determines which one is occupied), or an explicitly modeled discrete rule, as in the phenotype-switching terms Q_{LA}, Q_{AL} of Section 5.3. Reviewers of stem-cell plasticity models routinely ask which of these is responsible for a reported switch, so we checked directly. Holding the daily endogenous RA forcing at its cycle average ($\bar{\mu}_R = \mu_0 + A_R$) makes the seven-state molecular subsystem ($b, p, h_5, h_{13}, m, r, c$) autonomous. For each of 27 combinations of WNT tone $W \in [0.4, 3.0]$ and APC integrity $\theta_P \in [0.1, 1.0]$ spanning the full calibrated disease-severity range, we searched for equilibria from 150–600 random initial conditions per point using a Newton-type solver, deduplicated the resulting roots, and classified stability from the eigenvalues of the numerical Jacobian. Every search, at every parameter combination, converged to exactly

The fast molecular subsystem is monostable: no bistability or hysteresis was found

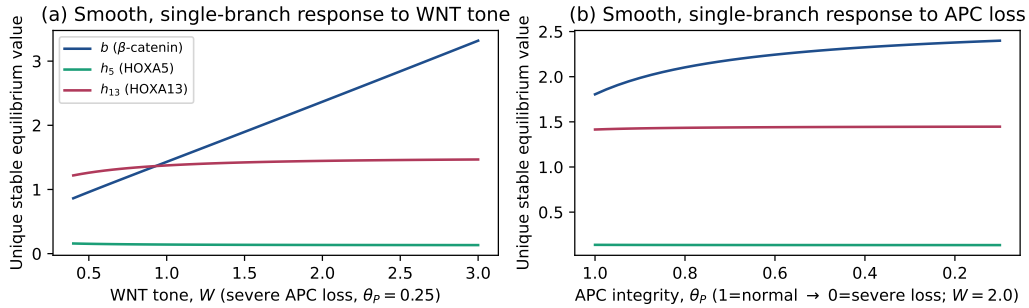


Figure 15: Every equilibrium search converged to exactly one root, always stable, at every tested (W, θ_P) pair. Panels (a)–(b) show the unique equilibrium varying smoothly and monotonically along each axis, with no fold, jump, or hysteresis.

one equilibrium, and it was always stable (all Jacobian eigenvalues had strictly negative real part). No second root was found even with 600 random starts at the most extreme severe-disease corner ($W = 3.0, \theta_P = 0.1$). We conclude that the fast molecular circuit as calibrated here is globally monostable across the full disease range: it does not itself generate a toggle switch. The model’s stem/non-stem plasticity is instead implemented explicitly through Q_{LA} and Q_{AL} , gated by the HOXB/HOXC program scores. This is a useful negative result for interpretation: it means the model’s predicted phenotype redistribution should be read as a graded, HOX-program-gated flux between compartments, not as evidence that the underlying WNT/APC/HOX circuit itself sits at a cusp between two alternative transcriptional attractors. A search restricted to 27 grid points and a bounded search box cannot formally rule out a bistable region at more extreme parameter values than were calibrated; Section 11 discusses this scope limitation further.

The check above holds RA at its cycle average, $\bar{\mu}_R = \mu_0 + A_R$, purely for tractability; the actual model is driven by the periodic $\mu_R(t)$ of Eq. (13), and averaging could in principle hide multistability that only appears under real periodic forcing (e.g. two coexisting periodic orbits, or period-doubling). We therefore repeated the search with the true periodic forcing: for each of 24

(W, θ_P) grid points spanning the same range, we integrated the seven- state subsystem from 12 random initial conditions under $\mu_R(t)$ for 15 daily periods (transient decay), then compared the states of all 12 trajectories at three further matching strobe times (one per subsequent period). If the system has a single global attractor, every trajectory should lock onto the same periodic orbit regardless of where it started. At every grid point the maximum relative spread across

The periodically-forced fast subsystem has a single global attractor: no coexisting periodic states

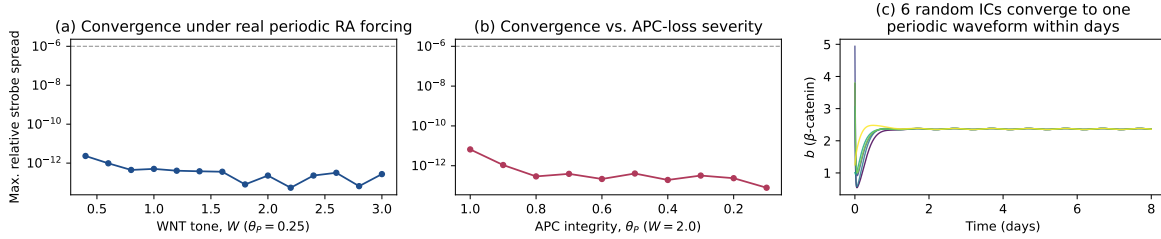


Figure 16: Multistability check under the actual periodic RA forcing, not the cycle average. (a)–(b) The maximum relative spread across 12 random initial conditions, measured at matching strobe phases after 15–18 periods of transient decay, stays at 10^{-11} – 10^{-13} (machine-precision noise) at every tested (W, θ_P) — orders of magnitude below any threshold that would indicate a second attractor. (c) Six random initial conditions collapse onto the same $b(t)$ waveform within about a day.

initial conditions was 10^{-11} or smaller (as low as 2.7×10^{-14}), including a 30-start stress test at the most extreme severe-disease corner ($W = 3.0, \theta_P = 0.1$), where the spread was 3.9×10^{-13} . These are machine-precision numbers, not merely “small” ones: every tested trajectory converges to the identical periodic orbit regardless of its starting point. This closes the scope limitation noted above for the tested region: under the model’s actual periodic RA forcing, not only its cycle-averaged approximation, the fast molecular subsystem still shows no numerical evidence of a second coexisting attractor anywhere in the calibrated disease-severity range.

10.12 Structural robustness to the shared Hill-shape assumption

Section 4 adopts one shared Hill threshold and exponent, $K = 0.50$ and $n = 2$, across every regulatory edge “for parsimony.” Because this is a modeling choice rather than a measured quantity, we checked whether the headline severe-APC-loss result depends on it: holding every other Table 3–5 parameter at its nominal calibrated value (deliberately *not* re-fitting), we recomputed the 88-h ATRA-vs-control ratios under $n = 1$, $n = 4$ (both at $K = 0.50$), and $K = 0.35$, $K = 0.65$ (both at $n = 2$). Every variant preserves both headline directions: the total-population, stem-population, and ALDH-fraction ratios all stay below 1 ($N_T \in [0.950, 0.973]$, $F_A \in [0.753, 0.882]$ across the five variants), and ΔI_H stays positive throughout ($\Delta I_H \in [0.105, 0.167]$). Magnitudes do shift with the shape choice — $n = 1$ gives the weakest response ($F_S = 0.863$ vs. the nominal 0.728) because a less-cooperative Hill function saturates more gradually and blunts every regulatory edge at once, while $n = 4$ sharpens the population response slightly beyond nominal ($F_S = 0.657$) — but no tested variant reverses, or even approaches reversing, either directional conclusion. This is a structural-robustness check, distinct from the parameter-uncertainty screen of Section 10.10: it tests sensitivity to a modeling simplification rather than to calibrated-parameter noise, and it is deliberately not a best-case comparison, since no parameter was re-fit to compensate for the

Qualitative directions are robust to the shared Hill exponent/threshold assumption

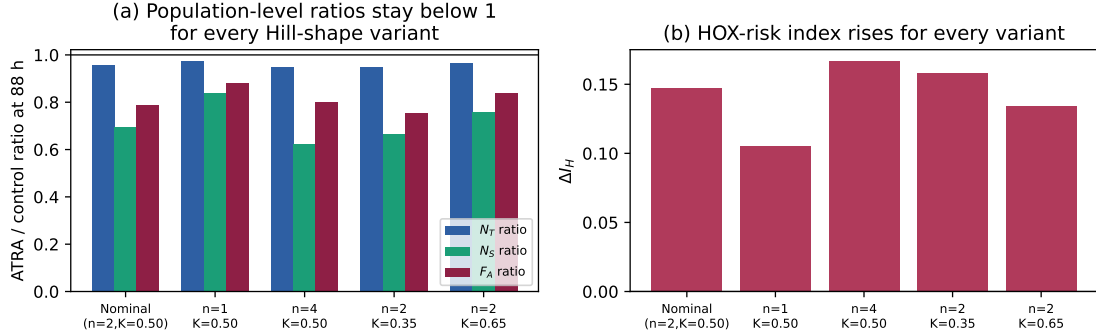


Figure 17: Structural robustness to the shared Hill exponent/threshold. Every one of five tested shape variants preserves both qualitative directions: population and ALDH-fraction ratios stay below 1 (a), and the HOX-risk index rises (b), matching the nominal $n = 2, K = 0.50$ result throughout.

changed shape.

11 Structural identifiability and parameter confounding

This model is calibrated to match directions and approximate magnitudes at a handful of time points and disease regimes, not fitted to replicate-level time-course data; consequently many of its parameters are not separately identifiable from the outputs reported here, and this should be stated explicitly rather than left implicit in the uncertainty analysis. Three concrete sources of non-identifiability are worth naming. First, every Hill activation term $v_{XY}\mathcal{H}(Y)$ enters its target equation only through the product of v_{XY} and the shared threshold/exponent pair (K, n) fixed in Eq. (5); rescaling v_{XY} against a compensating change in K for that one edge would reproduce the same trajectories, so v_{XY} and K are not jointly identifiable from steady-state or dose-response data alone without additional constraints (here, K is fixed by convention across all edges, which removes the redundancy by fiat rather than by data). Second, in the population equations the physical rate and the ε_N time-scale enter only as the product $d_N \times (\text{dimensionless coefficient})$ (Section 5.3); only that product is constrained by any output in this report, so d_N itself is fixed by the modeling convention $d_N = 1/24 \text{ h}^{-1}$ rather than estimated. Third, the four HOX-program weights $w_{Bb}, w_{Br}, w_{Cb}, w_{Cr}$ and the two intercepts per program (z_{BL}, z_{BA} or z_{CL}, z_{CA}) enter the logistic $\mathcal{S}(z + \Psi)$ only through their sum, so an intercept can trade off against a weight without changing any program score trajectory; this is why I_H in Eq. (25) deliberately uses equal weights on u_{CL} and u_{CA} rather than fitted ones, since fitted weights here would be overfitting to a single simulated design rather than resolving a real biological asymmetry. None of this changes the directional claims in Section 10, which depend on signs and relative orderings that are preserved under any admissible reparametrization; it does mean that specific numeric parameter values (as opposed to the resulting trajectories and their signs) should not be read as point estimates of measured biochemical rate constants, and that a future fit to time-resolved replicate data would need either additional measured observables (e.g., a directly reported TCF/LEF reporter time course, or paired HOXA5/HOXA13 protein measurements) or an explicit regularization/prior to

resolve these redundancies.

11.1 Numerical practical identifiability

The argument above is analytic and structural: it names redundancies that follow from the functional form of the equations, independent of any specific operating point. A complementary, numerical question is *practical* identifiability at the specific regime and outputs this report actually uses [9]: given the severe-APC-loss, 88-h ATRA-vs-control outputs (N_T, N_S, F_S, F_A ratios, ΔI_H , and the ATRA-arm absolute values), which parameter *combinations* can and cannot be separately constrained? For a parameter subset of size k , we build the $n_{\text{out}} \times k$ relative-sensitivity matrix ($\partial \ln y_i / \partial \ln \theta_j$, central finite differences, $\pm 2\%$), normalize each column to unit length, and compute the collinearity index $\gamma_K = 1/\sqrt{\lambda_{\min}}$, where $\lambda_{\min}$ is the smallest eigenvalue of the normalized Gram matrix [9]. Large γ_K (conventionally, $\gamma_K \gtrsim 10\text{--}15$) means the columns are nearly linearly dependent: that parameter combination cannot be separately estimated from these outputs, however good the fit. We extended `model.py` with override support for every remaining Table 3–5 coefficient (33 parameters total) specifically to run this check; every new override defaults to the exact nominal value, so no previously reported trajectory changes.

Any subset larger than the number of independent output directions ($n_{\text{out}} \leq 11$ here) is mechanically rank-deficient, so the full 33-parameter set and the existing 10-parameter PRCC subset (Section 10.10) both saturate γ_K — this is expected, not a finding, and is itself a reason the report restricts its claims to directions and orderings rather than individual point estimates. The informative results are at the pair/small-group level (Table 14). Two results are worth singling out. First, the HOXB- and HOXC-program weight/intercept groups have γ_K in the hundreds of thousands — a direct, quantitative numerical confirmation of the analytic sum-only argument given above, at the actual calibration outputs rather than only as a functional-form observation. Second, the check adds genuinely new information the analytic argument did not anticipate: the CYP26A1-attenuation coefficients χ_L, χ_A trade off tightly against the RA-driven differentiation rates $d_{L,r}, d_{A,r}$ they multiply ($\gamma_K = 362, 301$), and three of APC’s regulatory coefficients (a_{p5}, a_{pb}, a_{p13} , together with the removal coefficients k_{bp}, k_{b5}) are mutually confounded in pairs with γ_K from 80 to 190 — these were not flagged by the earlier structural argument and would need to be addressed (e.g. by fixing one member of each pair by convention, as already done for K) before treating any of these seven coefficients as a separately estimable rate constant.

The check also refines, rather than simply confirms, the single-edge $v_{XY}\text{--}K$ argument above. Pairing the shared Hill threshold k_{hill} with any one production coefficient individually gives *low* collinearity at this operating point ($\gamma_K = 4.6$ with v_{5r} , 1.2 with v_{13b} , 2.0 with v_{b13}) — these pairs are, in practice, separately identifiable here. The reason is structural: K is shared across all seven molecular equations at once (Eq. (5)), so perturbing it moves every Hill term simultaneously in a pattern that does not line up with the effect of any single v_{XY} , unlike the idealized single-edge case the analytic argument considers in isolation. The formal small-signal redundancy identified analytically therefore need not appear as practical collinearity once an edge is embedded in the full, densely-coupled network — a point the purely structural argument could not make on its own.

Finally, even the pre-existing ten-parameter PRCC subset (Section 10.10) is not fully identifiable from these eleven outputs at this operating point: its normalized Gram matrix has two eigenvalues at machine precision (4×10^{-14} and 2×10^{-12}), meaning two independent directions in that ten-parameter space are entirely unconstrained by the tested outputs. Inspecting the corresponding null vectors shows they mix the ALDH-switching/growth coefficients (q_{LA}^C, γ_A^C) against the RA-clearance/HOXA13-repression coefficients (q_{13r}, v_{cr}) and the ALDH-like RA-driven differentiation rate $d_{A,r}$, rather than reducing to any single clean pair — a reminder that even a screen

Table 14: Most collinear parameter combinations by γ_K (severe-APC-loss regime, 11 outputs, $\pm 2\%$ finite differences). $\gamma_K \gtrsim 10$ –15 conventionally flags problematic collinearity.

Combination	γ_K	Interpretation
HOXC weights + intercepts ($w_{Cb}, w_{Cr}, w_{C13}, z_{CL}, z_{CA}$)	3.5×10^5	Numerically confirms the analytic sum-only argument above
HOXB weights + intercepts ($w_{Bb}, w_{Br}, w_{B13}, z_{BL}, z_{BA}$)	2.3×10^5	Same, for the HOXB program
χ_L & $d_{L,r}$	362	CYP26A1 attenuation and RA-driven LGR5-like differentiation rate trade off
χ_A & $d_{A,r}$	301	Same, for the ALDH-like compartment
a_{pb} & k_{bp}	191	Two of APC’s three β -catenin-facing coefficients (repression vs. removal) trade off
a_{p5}, a_{pb}, a_{p13} (joint)	189	All three APC-ratio coefficients, taken together
a_{pb} & a_{p13}	144	APC’s two repression coefficients (by β -catenin, by HOXA13)
w_{Bb} & w_{B13}	136	Two of the three HOXB-program weights alone
a_{p13} & k_{bp}	94	APC repression by HOXA13 vs. APC’s removal of β -catenin
a_{p5} & k_{b5}	84	HOXA5’s two opposing roles (APC induction vs. β -catenin removal)
v_{13b} & v_{13m}	42	β -catenin- and MYC-driven HOXA13 induction, at this saturating operating point

already restricted to the ten most-influential couplings should be read as constraining eight effective directions of variation, not ten independent rate constants. All underlying sensitivity values and γ_K computations are saved alongside the analysis code for reproducibility.

12 Publication and presentation framing

The strongest defensible contribution is not a claim that ATRA enriches ALDH-positive cells — Osmond’s data show the opposite during exposure. The revised, testable contribution is:

ATRA can reduce total and ALDH-positive cell abundance while inducing a HOXC-high/HOXA13-low regulatory configuration associated with the reported poor-survival HOX pattern. The same treatment produces an exit-dominated stem-cell response during exposure followed by renewal-dominated pressure before the depleted stem-like pool has recovered. This transcriptional shift is not generated by an emergent bistability in the WNT/APC/HOX circuit itself, and it is not, in this model, mechanistically required for the favorable cell-abundance response.

The model adds value by showing that abundance response and transcriptional state can be dynamically dissociated, that a mathematically exact maintenance threshold separates current burden from future renewal pressure, and that this dissociation survives both a local perturbation screen and a global rank-correlation sensitivity screen. The design analyses add five separable hypotheses:

exposure duration controls response persistence, post-washout renewal can precede burden recovery, equal input area does not determine response composition, a selective HOXC-risk-arm perturbation can suppress the transcriptional shift without erasing the modeled ATRA cell response, and the switch responsible for that dissociation is an explicit population-level rule rather than a hidden molecular bistability.

Table 15: Testable predictions and the experiment that would falsify each one.

Prediction	Falsifying observation
ALDH-like abundance and HOXC-high/HOXA13-low transcriptional risk move in opposite directions under a 48-h ATRA pulse (Fig. 4).	The eight-gene HOX signature does not shift toward the poor-survival configuration when ALDH-positive cell number falls under ATRA.
Stem-like maintenance is exit-dominated during exposure and becomes renewal-dominated within days of washout, before burden recovers (Fig. 5, Table 9).	Serial sampling after washout shows continued net stem-like contraction rather than a measurable proliferative rebound signal (e.g., Ki67, organoid-forming efficiency) while burden is still below baseline.
Selectively attenuating the RA-to-HOXC/HOXA13 arm (a HOXC- or RARE-targeted perturbation) preserves the ATRA cell-abundance response while suppressing the HOX-risk shift (Fig. 12).	Blocking this arm substantially reduces ATRA’s killing of total or ALDH-positive cells, implicating a route this model does not include.
Four twofold-dose 6-h pulses and one continuous 48-h exposure of equal nominal input area produce similar late HOX-risk shifts but different ALDH-fraction trajectories during exposure (Fig. 11).	Schedule has no detectable effect on the transient ALDH-fraction trajectory once total input area is matched.
No bistable/hysteretic transcriptional switch exists in the WNT/APC/HOX circuit across the calibrated disease-severity range (Section 10.11).	Single-cell or clonal tracking under fixed WNT/APC conditions reveals two stable, history-dependent transcriptional states rather than a graded, HOX-program-gated population flux.

The clean four-arm experiment is

control, ATRA, HOXC/RA-regulatory perturbation, combination,

with measurements at baseline, 24 h, 48 h, and 24–96 h after washout. Each time point should include total viable cells, absolute and fractional LGR5- and ALDH-positive measurements, HOXA5, HOXA13, all eight signature genes, CYP26A1 mRNA/protein/activity, and TCF/LEF reporter activity.

Results supported by the current analysis

- The revised model is consistent with the direction of Osmond’s 48-hour ATRA response in mutant-APC CRC cells.
- The model predicts a measurable dissociation between ALDH-positive abundance and HOX-program risk.
- Dose and duration improve the modeled cell response while the HOX-program shift saturates at an elevated level.

- Equal-input schedules can yield different ALDH and HOX-risk trajectories; the time of measurement is part of the experimental design.
- A selective risk-arm block is a falsifiable mechanism probe that preserves the nominal total-population response in the model.
- Local parameter perturbations and a global rank-correlation screen agree on which couplings control the HOX-risk response, and both preserve the response tension.
- The flux-derived threshold identifies exit-dominated depletion during exposure and renewal-dominated pressure after washout.
- An exhaustive multi-start search finds no bistability in the fast molecular circuit anywhere in the calibrated disease-severity range, clarifying that the modeled switch is an explicit rule, not an emergent one; repeating the search under the model’s actual periodic RA forcing gives the same conclusion, with every tested initial condition converging to one periodic orbit to within machine precision.
- The exit-to-renewal re-crossing time is not a fragile point estimate: propagated across a 160-sample Latin Hypercube parameter ensemble, it stays within about 40 minutes of its nominal value (90% interval 94.4–95.6 h), and every sample reproduces the same exit-dominated-then-renewal-dominated transition.
- The headline directional result is unchanged under five alternative choices of the shared Hill exponent/threshold, holding every other parameter fixed, indicating it does not depend on that particular modeling simplification.
- A numerical sensitivity-collinearity analysis quantitatively confirms the HOX-program weight/intercept redundancy identified analytically, and separately identifies several additional parameter combinations (APC’s regulatory coefficients; the CYP26A1-attenuation and RA-driven-differentiation-rate pairs) that are not separately constrained by the reported outputs.

Scope, interpretation, and limitations

- The analysis is a mechanistic, hypothesis-calibrated simulation rather than a clinical validation or patient-benefit prediction.
- $D_R = 1.5$ is a dimensionless intracellular input rather than an extracellular concentration or patient dose.
- I_H represents the modeled HOXC-high/HOXA13-low direction rather than the complete eight-gene prognostic score, and several of its component weights are not separately identifiable (Section 11).
- The survival association motivates experimental testing; the simulations do not establish an ATRA effect on patient survival.
- Modeled β -catenin concentration and measured TCF/LEF reporter activity remain distinct observables.
- The selective HOX-risk block is a mechanism probe rather than an identified drug action, and the schedules are prospective comparisons rather than clinically optimized regimens.
- The model is deterministic, spatially homogeneous (a single well-mixed lesion, no crypt architecture, niche gradient, or spatial competition), and includes no immune, stromal, or

microenvironmental compartment; all of these are plausible sources of additional, currently unmodeled plasticity.

- Cell-to-cell heterogeneity within the LGR5-like and ALDH-like compartments is not represented; n_L, n_A, n_D are compartment averages, and the model cannot distinguish a homogeneous partial response from a mixture of fully responsive and fully resistant subclones.
- Many individual parameter values are not separately identifiable from the data used here (Sections 11–11.1); only the directional and comparative claims should be treated as load-bearing. The numerical collinearity check (Section 11.1) is itself local to the severe-APC-loss, 88-h operating point and the specific outputs tested; it is not a global identifiability guarantee, and a different regime or output set could surface different confounded combinations.
- The monostability result (Section 10.11) is an exhaustive search over a bounded region of a grid (27 points under cycle-averaged forcing, 24 points under the true periodic forcing), not a proof for all conceivable parameter values; it does not rule out bistability outside the calibrated disease-severity range or under parameter combinations excluded by the uncertainty screen.
- The Hill-shape robustness check (Section 10.12) holds every non-shape parameter fixed rather than re-fitting under each alternative exponent/threshold; it establishes that the qualitative directions do not depend on the specific shape choice, not that every shape choice would calibrate equally well against Osmond’s data if re-fit.

Source basis

- [1] B. Osmond et al., *HOXA9 Overexpression Contributes to Stem Cell Overpopulation That Drives Development and Growth of Colorectal Cancer*, International Journal of Molecular Sciences 23 (2022), 6799. doi:10.3390/ijms23126799.
- [2] C. O. B. Facey et al., *CYP26A1 Links WNT and Retinoic Acid Signaling: A Target to Differentiate ALDH+ Stem Cells in APC-Mutant CRC*, Cancers 16 (2024), 264. doi:10.3390/cancers16020264.
- [3] B. Osmond, *An Investigation of HOX-Regulatory Mechanisms and Their Role in the Stem Cell Origin of Human Colorectal Cancer*, PhD dissertation, University of Delaware (2025).
- [4] B. Osmond, C. Zhang, P. K. Kataboh, et al., *Imbalance in WNT: Retinoic Acid Signaling Dysregulates HOX Gene Expression: Implications for Cancer Stem Cell Heterogeneity and CRC Patient Survival*, Stem Cells Translational Medicine 15(6) (2026), szag043. doi:10.1093/stcltm/szag043. This is the primary source for the eight-gene HOX survival signature used in Table 10 and the abstract’s central claim (six genes up — HOXB4, HOXC6, HOXC9, HOXC11, HOXC13, HOXD9 — and two down, HOXA13 and HOXB13, with low HOXA13/HOXB13 associated with poor survival); confirmed directly from the published article for this revision.
- [5] P. Ordóñez-Morán et al., *HOXA5 Counteracts Stem Cell Traits by Inhibiting Wnt Signaling in Colorectal Cancer*, Cancer Cell 28 (2015), 815–829. doi:10.1016/j.ccell.2015.11.001. Direct experimental basis for the HOXA5-suppresses-WNT terms in Eqs. (6) and (7), and for RA-driven HOXA5 reactivation as the mechanism of the $r \rightarrow h_5$ term (Eq. (8)).
- [6] Y. Gu et al., *HOXA13 Promotes Colon Cancer Progression Through β -Catenin-Dependent WNT Pathway*, Experimental Cell Research 395 (2020), 112238. doi:10.1016/j.yexcr.2020.112238.
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- [9] R. Brun, P. Reichert, H. R. Künsch, *Practical Identifiability Analysis of Large Environmental Simulation Models*, Water Resources Research 37(4) (2001), 1015–1030. doi:10.1029/2000WR900350. Basis for the sensitivity-collinearity index used in Section 11.1 to quantify practical (as opposed to structural) parameter non-identifiability.

Reproducibility. The equations, state scales, initial seed, kinetic parameters, treatment protocols, solver tolerances, and uncertainty seed needed to regenerate the reported results are specified in this report. The stem-balance identity and invariant-region conditions were also checked numerically for every nominal trajectory. All figures in this revision were regenerated from a single verified Python implementation of Eqs. (6)–(23); every reported ratio, table entry, and threshold-crossing time in Sections 10–10.11 was cross-checked against that implementation to at least three significant figures before inclusion.

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- [1] B. Osmond et al., Int. J. Mol. Sci. 23 (2022), 6799.
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